Venkata Lakshmi Parimala Pasupuleti

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SUMMARY

Backend AI engineer with 6+ years of experience building production LLM systems, multi-agent orchestration, and distributed backend services. Built LeaseI, an 8-agent LangGraph platform with HITL escalation and eval harnesses (93.6% precision across 1,240 test cases), and AgentProbe, an open-source LLM evaluation toolkit with provider routing across five LLMs. Currently integrating Anthropic Claude into financial services at Intuit -- prompt design, structured-output schemas, guardrails, and LLM-as-judge regression tests. Strong in Python/FastAPI, async/distributed systems, idempotent processing, and fine-grained authorization. Daily Claude Code user.

CORE COMPETENCIES

Multi-Agent Orchestration (LangGraph) | Python / FastAPI | LLM Eval Harnesses | LLM-as-Judge | Prompt Engineering & Caching | Model Routing | Human-in-the-Loop (HITL) | Distributed Systems | Async Processing (RabbitMQ) | Idempotency & Retry Logic | OAuth / RBAC / Fine-Grained Authorization | AWS (Lambda, S3, RDS, SQS, API Gateway) | GCP (Vertex AI) | Docker / Kubernetes | GitHub Actions CI/CD | Browser Automation (Playwright) | Prometheus / OpenTelemetry | Java / Spring Boot | React / TypeScript

EXPERIENCE

Intuit

Jun 2026 -- Present

Software Engineer -- Virginia, USA

- Integrate Anthropic Claude into Python/FastAPI financial services -- **structured-output schemas**, **prompt caching**, and retrieval-grounded responses so Claude answers stay consistent and traceable.
- Build **LLM-as-judge eval harnesses** with golden datasets that run in CI, catching hallucinations and quality regressions before release.
- Implement **guardrails** around LLM calls: input validation, sensitive-data handling, and fallbacks when model output fails checks.
- Build ML pipelines (scikit-learn, XGBoost) from feature engineering through prediction APIs; ship services via Docker and GitHub Actions CI/CD on AWS.
- Instrument services with **Prometheus and OpenTelemetry** for production observability across p50/p99 latency and error states.

Entrata -- Apartment & Affordable Housing SaaS

Oct 2020 -- Jul 2024

Senior Software Engineer -- Multi-tenant platform, 100K+ monthly active users

- Built **RabbitMQ-based async processing** for rent updates across thousands of tenant records -- **idempotency guarantees and retry logic** preventing double-processing in a multi-tenant production system.
- Designed and maintained **distributed backend services** (Python, Java, PostgreSQL) handling eligibility, financial calculations, and document workflows with 2-3x traffic spikes at month-end billing cycles.
- Implemented **fine-grained authorization**: role-based access control and multi-tenant data isolation for property managers, leasing staff, and residents across 100K+ MAU.
- Built **Libsodium-encrypted data pipelines** covering 100% of resident PHI; led AWS S3 document migration with metadata validation and cutover testing.
- Optimized 200+ SQL queries via execution plan analysis -- 30% average performance improvement; delivered 20+ features across 120+ Agile sprints; reviewed 300+ pull requests.

Entrata India

Sep 2018 -- Oct 2020

Junior Software Engineer -- Pune, India

- Built property management backend modules (Python, Java, PostgreSQL) with async data import pipelines processing 1.6M HUD/Tax Credit records.

- Led AWS S3 cloud migration from legacy server storage; automated 56 state-specific housing certification documents via merge-field pipelines.

PROJECTS

Leasel -- 8-Agent LangGraph Platform with HITL

Python · LangGraph · FastAPI · Multi-Agent · Eval CI

Multi-agent orchestration platform for housing compliance: triage, retrieval, reasoning, validation, and HITL escalation agents with real conditional edges. **Agent runtime**: context management, confidence-based routing, and handoff to human when confidence falls below threshold -- agent never auto-resolves when escalation is warranted. Eval harness across 1,240 test cases: 93.6% precision, 87.9% F1 via LLM-as-judge CI.

github.com/venkatalakshmiparimala/Lease_complaince_rag

AgentProbe -- LLM Evaluation & Provider Routing Toolkit

Python · FastAPI · LangGraph · Model Routing · LLM-as-Judge

15-endpoint FastAPI backend with **model routing** across five LLM providers (Anthropic, OpenAI, Azure OpenAI, Groq, Ollama). Scores outputs on factual accuracy, hallucination rate, retrieval quality, completeness, and compliance using LLM-as-judge. CI regression against golden datasets; 3 LangGraph agents (Orchestrator, RAG Tracer, Auto Fix). Runs 100% locally at zero cost.

github.com/venkatalakshmiparimala/AgentProbe

AI on FHIR -- Clinical NLP Backend

Python · Flask · OAuth · RBAC · Docker · FHIR REST APIs

Backend API translating natural-language clinical questions into FHIR REST calls using LLMs and BERT. **Fine-grained authorization**: role-based access control with HIPAA-style data isolation across user tiers. Containerized with Docker; ~60% reduction in manual clinical lookup time.

github.com/venkatalakshmiparimala/AI_on_FHIR

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia — May 2026 | GPA 3.87

B.Tech Computer Science & Engineering — JNTU Anantapur — 2017